



**Internship Report
On
IOT BASED SMART IRRIGATION SYSTEM**

Submitted to the Savitribai Phule Pune university, Pune in partial fulfillment of the requirements for the award of the degree B.E. Computer Science

**Submitted by
Divyaamshu Verma**

**Under the Guidance of
Mr. Rajesh Bharati**

DEPARTMENT OF COMPUTER SCIENCE

Dr. D.Y. Patil Institute of Technology, Pimpri, Pune-18

2021-22

Dr. D. Y. Patil Unitech Society's
Dr. D.Y. Patil Institute of Technology,Pimpri,Pune-18

CERTIFICATE

This is to certify that the internship report entitled

“IOT based smart irrigation system ”

Submitted by

D i v y a a m s h u

V e r m a

Roll No.:

TCOD01

Exam Seat No : T190244266

is a bonafide work carried out under the supervision of Mr. Rajesh Bharati and it is submitted towards the partial fulfillment of the requirement of Savitribai Phule Pune University, Pune for the award of the degree of Bachelor of Engineering (Computer Science) in the academic year 2021-2022.

Mr. Rajesh Bharati
(Internship Guide)

Dr. Pramod D. Patil
(Principal)

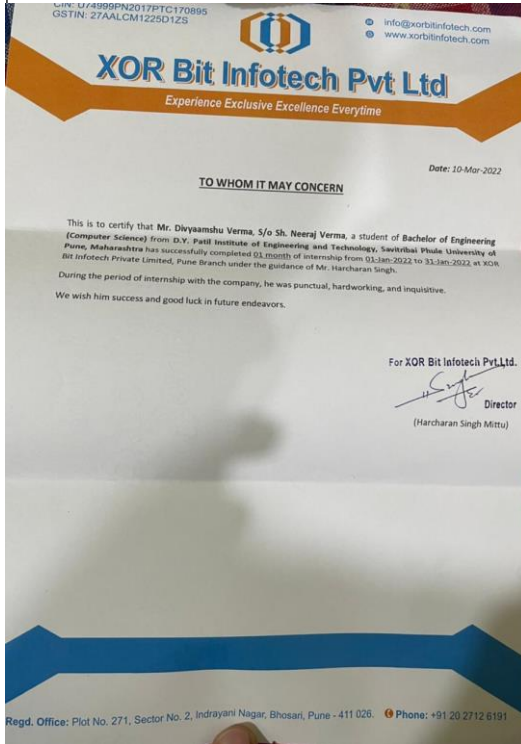
Seal/Stamp of the College

Examiner 1 :

Date:

Examiner 2 :

Internship completion certificate



Offer Letter

CIN: U74999PN2017PTC170895
GSTIN: 27AALCM1225D1ZS

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XOR Bit Infotech Pvt Ltd
Experience Exclusive Excellence Everytime

Confidential

Offer Letter

Date: 22-Dec-2022

Mr. Divyaamshu Verma,
S/o Sh. Neeraj Verma
House No. 550, Sector 2
Panchkula, Haryana - 134 112.

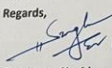
Dear Divyaamshu,

With reference to your interest, followed by further meetings, it is a very pleasant feeling to finally let you know that you are now going to be among the exclusive folks at XOR Bit Infotech. Through various levels of engagements, interviews, and discussions we have had so far, we are pleased to offer you internship to work with us as **Trainee Engineer (IoT Services)**. The duration of the engagement will be minimum 30 days and may extend depending on the work assigned to you.

As explained to you, you will represent and work as an independent resource under no commission or salary or any other form of compensation. Following are the agreed terms of engagement (the "Offer"):

Location	Indrayani Nagar, Pune	Probation Period	Not Applicable
Latest Reporting Date	01-Jan-2022	Notice Period	Not Applicable
Reporting To	Harcharan Singh	Offer Valid Until Date	31-Dec-2021
Documents Required	As per Annexure A		

Welcome on board and looking for long term beneficial relationship. All the best!
For XOR Bit Infotech Pvt Ltd.

Regards,
 Director
(Harcharan Singh)



Regd. Office: Plot No. 271, Sector No. 2, Indrayani Nagar, Bhosari, Pune - 411 026. Phone: +91 20 2712 61

Internship Place Details

1. Company Name – **XOR Bit Infotech Pvt. Ltd**

2. **Location** – **Pune, India**

3. Company Background-

XOR Bit Infotech is an IT service providing company, which offers cloud services, IDC consultancy, IT training, software development, Internet of things, Artificial intelligence & Machine learning services. IT follows the motto: experience exclusive excellence every time.

4. **Mission**– Our mission is to facilitate our clients with cost effective solutions built upon robust and automated infrastructure by the most competent and trained human resources.

5. **Personal Observations** – The company was very well structured into levels where the top positions were occupied by highly skilled professionals and each and every code change underwent end to end review which helped increase the efficiency of implementation of the project. The projects were taken over by the firm and the tech stack was the industry standard, thus attracting very high potential clients.

Acknowledgment

Firstly, I would like to thank Mr. Harcharan Singh Mittu for giving me the opportunity to do an internship within the organization.

I would also like all the people that worked along with me with their patience and openness. They created an enjoyable working environment. It is indeed with a great sense of pleasure and immense sense of gratitude that I acknowledge the help of these individuals.

I am extremely grateful to my guide Mr. Rajesh Bharati and the department staff members and friends who helped me in successful completion of this internship.

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Introduction

I had applied for the internship at Company's posting on [LinkedIn](#). After applying I had a couple of interviews to get the internship which were:

1. Telephonic Interview – The telephonic interview was basically a short introductory call to test my understanding on the tech stack and help me understand the company culture and work, their past projects and a brief about their past projects. After they were satisfied with my understanding of the tech stack, they waved a green flag and I was passed on to the next set of interview processes.
2. Technical Interview – The technical interview was focused on testing my foundational understanding of IOT, my past work experiences and some advanced C++ problems to test my problem solving and development skills. It was a 45-minute interview round after which I was selected for the internship.

IOT based smart agriculture controlling system:

This project presents a new design of water pump control for the development of a smart irrigation system linked with SMS.

Wastage of water in the current scenario, merely due to overflowing tanks, which is not affordable. Conventional water tanks can neither monitor nor control the water level in a tank, leading to large amounts of water wastage. Some other technologies had certain drawbacks in some or the other way. The need to remove these short-comings and providing an efficient and economical solution has been the main focus of this project.

Our project is for water level monitoring as well as controlling with IOT and GSM modules. The IOT platform we are using is Arduino which is an open source. A Float-level sensor is placed on the surface of the tank to sense the water level. We can monitor the tank manually using an on/off system provided to the user controlled through a GSM module via text message. Index Terms: IoT, GSM module, ultrasonic sensor, Arduino.

Problem Statement

Water is always a crucial part of everyday life. Due to the global environmental situation, water management and conservation is vital for human survival. In recent times, there were huge needs for consumer-based humanitarian projects that could be rapidly developed using Internet of Things (IoT) technology. Our proposed system measures water level in real-time and helps the user to monitor the water tank remotely using a mobile device via text message. It helps farmers which are not present on fields to remotely check the status of their water tank and provides access to them to operate the motor pump remotely at any time.

Objectives

- The main objective of the project was to create an IOT based system which is easy to access and operate.
- To help farmers check electricity status on fields and water level in their tanks remotely
- Also to provide them a way to remotely control the water pump present on their fields using their mobile phone

Motivation

Wastage of water in the current scenario, merely due to overflowing tanks, which is not affordable. Conventional water tanks can neither monitor nor control the water level in a tank, leading to large amounts of water wastage. Some other technologies had certain drawbacks in some or the other way. The need of removal of these short-comings and providing an efficient and economical solution has been the main focus of this project.

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Scope

The scope of the project was-

- Improved productivity of staff and reduced human labor
- Efficient operation management
- Better use of resources and assets
- Cost-effective operation
- Time efficient IOT model
- Remote controlling using mobile phone
- Improved customer service and retention
- Easy to operate IOT model via sms

System Analysis

The system requirements were –

Hardware Requirements:

- Arduino Uno - Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online.
- GSM 800L Module - The Arduino GSM shield allows an Arduino board to connect to the internet, send and receive SMS, and make voice calls using the GSM library.
- DC Pump - DC powered pumps use direct current from motor, battery, or solar power to move fluid in a variety of ways. Motorized pumps typically operate on 6, 12, 24, or 32 volts of DC power.
- Float Level Sensor - Float level sensors are continuous level sensors featuring a magnetic float that rises and falls as liquid levels change. The movement of the float creates a magnetic field that actuates a hermetically sealed reed switch located in the stem of the level sensor, triggering the switch to open or close.
- LCD Display and I2C Module - I2C_LCD is an easy-to-use display module; it can make displays easier. Using it can reduce the difficulty of making, so that makers can focus on the core of the work.
- Relay Module (5V 1Channel) - The relay module is an electrically operated switch that can be turned on or off to let current flow through or not.
- Breadboard - A breadboard is used to build and test circuits quickly before finalizing any circuit design.
- Jump Wire - A jump wire is an electrical wire, or group of them in a cable, with a connector or pin at each which is normally used to interconnect the components of a breadboard.
- Step Down Module (Lm2596) - is a step-down(buck) switching regulator, capable of driving a 3-A load with excellent line and load regulation. These devices are available in fixed output voltages of 3.3 V, 5 V, 12 V, and an adjustable output version.

Software Requirements:

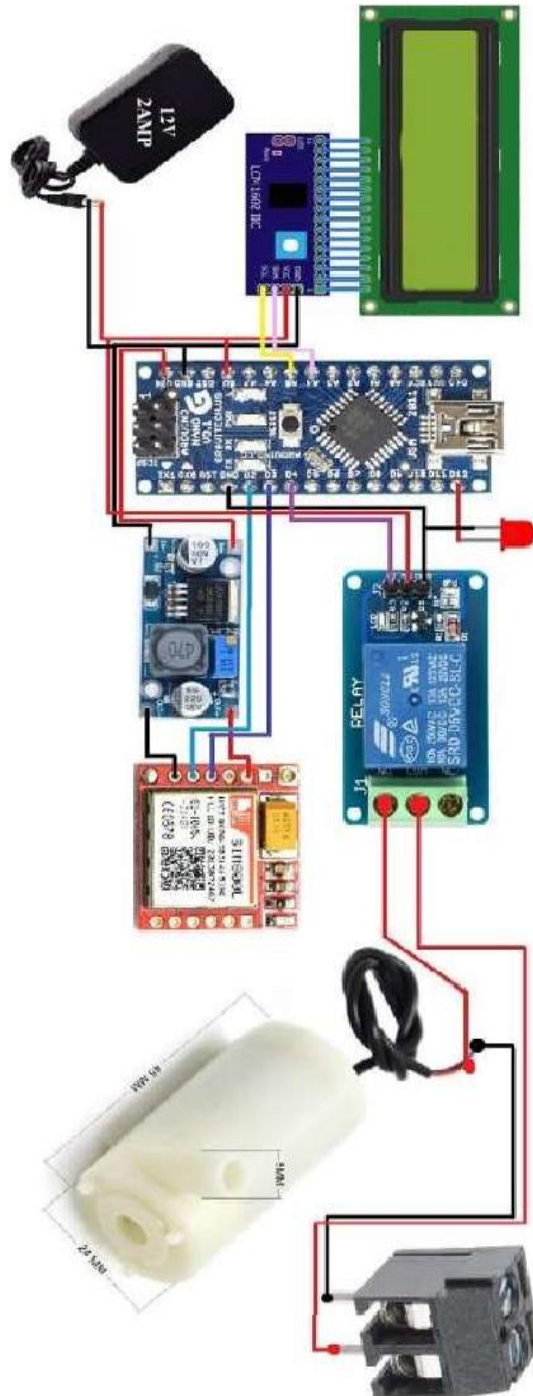
- Arduino operating system - The Arduino Uno is programmed using the Arduino Software (IDE), the Integrated Development Environment common to all the boards and running both online and offline.
- GSM - This module is able to receive serial data from radiation monitoring devices such as survey meter or area monitor and transmit the data as text SMS to a host server.

System Design:

This module is composed of a Float-level sensor, LCD Display, Arduino UNO microcontroller, SIM800L GSM Shield module, Step-Down Converter and 240VAC relay switch.

For monitoring, a Float-level sensor was used because of its capability to probe inside a water tank. The Float-level sensor is connected to the Arduino UNO microcontroller and the signal sent by the sensor will be interpreted and processed by the microcontroller to evaluate the level of water in the tank. Based on the uploaded code, if the level of water requires filling the tank, then the Arduino microcontroller will send a signal to turn on the solid-state relay switch where the water pump is connected. As a control measure, the same process avoids the overflow of the water tank or before it gets empty. If the water is already at full level, the Arduino microcontroller will send a signal to the solid-state relay switch to turn off and when the water level is at a minimum level it will turn on. The microcontroller sends a signal to the solid-state relay switch where the motorized electric valve is connected to close and open. The micro-controller will also activate the GSM module that is connected to send data to the server to report the status of the water tank and current activities of the water pump. The data will be sent through SMS which is believed to be the most practical and economical way of communication. The monitoring and controlling systems are also allowed to receive SMS to override the built-in automatic controlling system of this module.

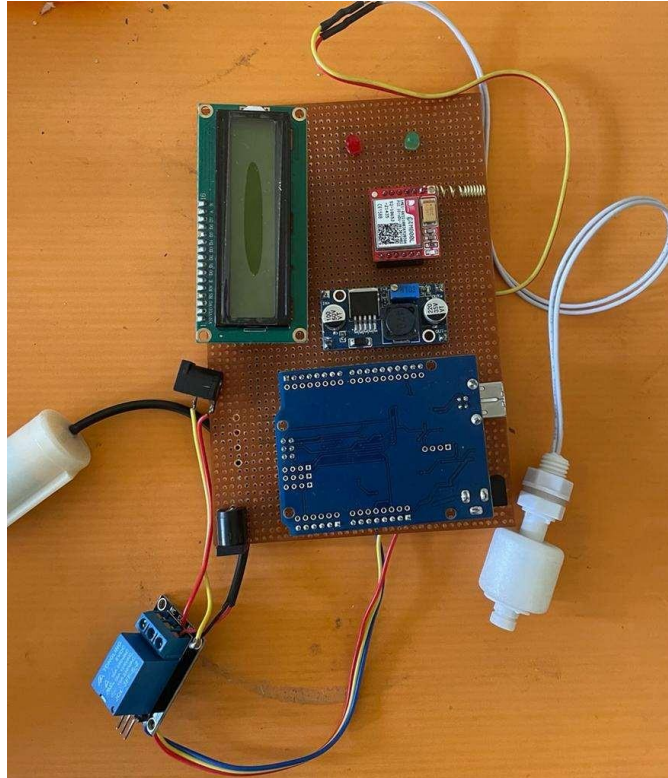
Circuit Block Diagram:



Algorithm for project:

1. Start
2. Check water level in water tank
3. If below threshold, the water pump will start automatically and send SMS to registered mobile number
4. If the user wants to stop the pump, they can simply do it by sending via SMS.
5. If the water reaches the maximum threshold, the pump will stop automatically and send a SMS to the user that the water tank is full.
6. Stop

PROJECT MODEL:



EXAMPLE CODE:

```
if ((number[0] == str[0] && number[1] == str[1] && number[2] == str[2] && number[3] == str[3]
    && number[4] == str[4] && number[5] == str[5] && number[6] == str[6] && number[7] == str[7]
    && number[8] == str[8] && number[9] == str[9] && number[10] == str[10] && number[11] ==
str[11]))
{
    gsm.println("ATH");
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Unknown number");
    lcd.setCursor(0, 1);
    lcd.print("Call Rejected.");
    wr_call_val = 1;
    EEPROM.write(wr_call_add, wr_call_val);
    delay(2000);
    resetFunc();
}

if (EEPROM.read(address) == 0)
{
    serialEvent();
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("System Standby");
    lcd.setCursor(0, 1);
    lcd.print("Status: OFF");
    delay(1000);
}
```

System Implementation:

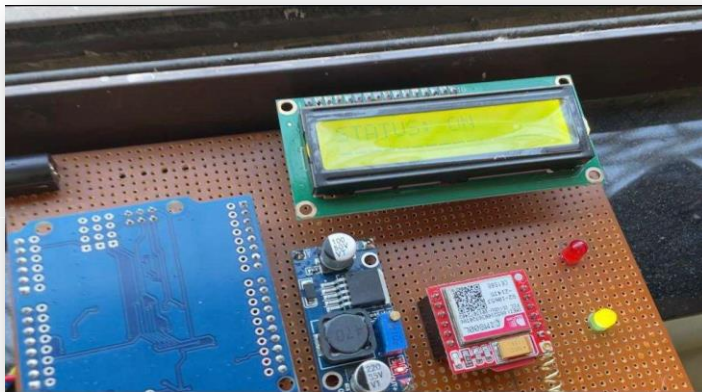
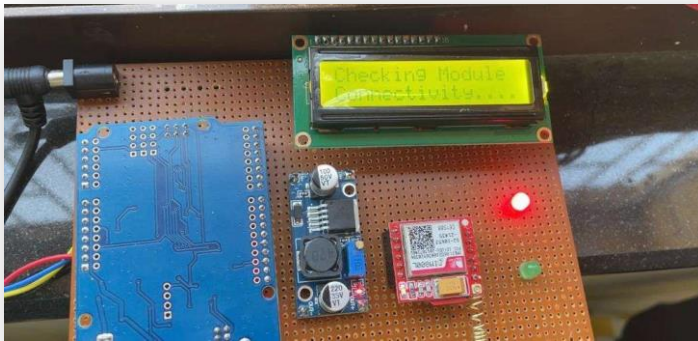
- For this project, we have used Arduino UNO as a microcontroller. The values of maximum and minimum levels are obtained by Arduino using the ESP module. These values are set in the code written for Arduino UNO. The current level of water is obtained from the Float-level sensor. Depending upon these values, the motor is turned ON / OFF. Depending on the water levels the status of the motor will be automatically controlled. If water level is in between both the levels, then the user can exercise control of the motor using his mobile device via text message. ON and OFF have been provided for the same. The height of the tank is to be set once in Arduino. This height shall be used to determine the percentage of water. Calculations of the current water level will be done with this.

System Features:

1. Remote control of the water pump using SMS.
2. Notification on SMS alerts when motor status changes.
3. Setup schedule for switching the pump on/off.
4. Module operates over GSM.
5. Easy to install.
6. Sim card insertion, on board GSM module.
7. Can be set up with an existing pump.

Results:

So as shown in previous slides, I have successfully finished the IOT smart irrigation system project assigned to me by making a properly working IOT model with an executable Arduino code implemented in it which checks the water level in the tank and controls the motor pump via SMS through mobile phone.



Conclusion

So, by doing this IOT based smart irrigation system project, I have successfully finished my one month internship at XOR Bit infotech Pvt. Ltd. On 31th of January.

Attendance Record

Week	Work Duration	Work
1 st week	3 hours / day	Basic Training in IOT by studying and working on existing projects
2 nd week	4 hours / day	Creating own IOT Projects as per the instructions
3 rd week	4 hours / day	Working on IOT Project and making changes in it as per requirement
4 th week	4 hours / day	Completing the project by making a fully functional IOT model

References:

For company information:

1. XOR Bit Infotech Website: <http://www.xorbitinfotech.com/>
2. XOR Bit Infotech LinkedIn Page: <https://in.linkedin.com/company/xorbitinfotech>

